

MANIKANTA PUDI

Hyderabad | pudimanikanta3@gmail.com | +91 9959540601 | manikanta.tech | linkedin.com/in/manikanta3 | github.com/Mani9kanta3

PROFESSIONAL SUMMARY

Entry-level AI Engineer who builds retrieval and agent systems, then measures them honestly. Built a LangGraph SQL agent with 121 tests proving unsafe SQL cannot execute, and a hybrid RAG assistant that filters eligibility in SQL before vector search, raising threshold accuracy from 0.83 to 0.92. Applied Python and SQL to 6.7M+ records over 6 months of paid analyst work.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI & Agents: LangGraph, MCP, LangChain, RAG, Hybrid Retrieval, LLM-as-Judge Evaluation, RAGAS, Groq, Gemini, HuggingFace

Retrieval & Vector Stores: ChromaDB, Embeddings (bge-small-en-v1.5), Cross-Encoder Reranking, Retrieval Evaluation, Abstention

Machine Learning: scikit-learn, XGBoost, Random Forest, Pandas, NumPy, Feature Engineering

Backend & Databases: FastAPI, Flask, Django REST Framework, REST API Design, Pydantic, PostgreSQL, SQLAlchemy ORM, sqlglot

DevOps & Tools: Docker, Git, Render, Vercel, Langfuse, Pytest, Linux

PROFESSIONAL EXPERIENCE

Data Science Intern | Infosys Springboard

December 2025 - February 2026

Remote, India

- Developed an AI packaging recommendation engine across 25 materials and 13 categories, returning suitability, cost and CO2 predictions in real time.
- Trained Random Forest and XGBoost on 2,275 samples with 15 engineered features, reaching 0.97+ R-squared on suitability and 0.98+ on CO2 impact.
- Served the models through a Flask REST API with joblib loading, deployed on Render with managed PostgreSQL.

Data Analyst Intern | Techmatics Solution

January 2025 - June 2025

Hyderabad, India

- Analyzed 6.7M+ retail transactions in Pandas, finding that 20% discounting reduced net revenue despite higher volume.
- Wrote multi-table JOINS, CTEs and window functions against PostgreSQL, reducing query time and removing manual preparation from reports.
- Built reusable Python functions for data validation, cleaning and aggregation used across reporting pipelines.

PROJECTS

Scholarship Eligibility Assistant - Hybrid RAG System | [GitHub](#) | [Live App](#)

August 2026

Python, FastAPI, LangChain, ChromaDB, PostgreSQL, Groq, RAGAS, bge-small-en-v1.5, React, Render

Eligibility Q&A over 44 schemes. "Minimum 80% marks" and "minimum 75%" embed almost identically, so semantic search returns the wrong threshold.

- Found that plain semantic retrieval surfaced 54 schemes with 81% ineligible, so moved eligibility rules into typed SQL columns filtered before retrieval, raising threshold accuracy from 0.83 to 0.92.
- Added regex and LLM-as-judge grounding that refuses unsupported answers, reducing invented figures from 7.5% to zero, with judge agreement of 0.85 across 48 tests.
- Dropped the cross-encoder reranker after two runs scored 0.88 and 0.93, a swing inside noise at 40 questions.

SQL Analyst Agent - Text-to-SQL with Self-Repair | [GitHub](#) | [Live App](#)

September 2026

Python 3.12, LangGraph, MCP, FastAPI, PostgreSQL 17, sqlglot, Groq, Langfuse, React 19, Render

Agent that answers plain-English questions over a PostgreSQL database, returning the answer and the SQL that produced it.

- Mapped PostgreSQL error codes to distinct repair strategies, so an unknown column widens schema retrieval instead of retrying blindly.
- Enforced a read-only session, 5-second timeout, forced LIMIT of 200 and sqlglot rejection of non-SELECT, proven by 121 tests not by sampling.
- Measured the loop against a single-shot baseline on 40 hand-written questions, refusing all 10 unanswerable ones correctly.

CERTIFICATIONS & ACHIEVEMENTS

- Codegnan - Full Stack Python Development Certification (Ongoing, since June 2026).**
- Research Publication - Off-Road Terrain Identification using YOLOv8, Journal of Electrical Systems (December 2023).**
- SQL (Intermediate) Certification - HackerRank (March 2025).**

EDUCATION

Bachelor of Engineering in Computer Science (Artificial Intelligence & Machine Learning)

2021 - 2025

Chandigarh University, Mohali | CGPA: 7.98/10

Final Year Project: Traffic Sign Detection using YOLOv8, 92% classification accuracy.